Date

Expt. No. 04 Page No. 33

AIM OF THE EXPERIMENT

To design an Astable Multivibrator using a 555 timer and measure it's frequency and waveform characteristics.

OBJECTIVES

i) To construct an astable multivibrator using 555 timer.
ii) To observe the output waveform and capacitor voltage waveform.
iii) To determine the time periods and duty cycle.
iv) To measure the output frequency.

COMPONENTS REQUIRED

Sl no.	Name of the components	Specification	Quantity
1	IC 555 timer	NE555 timer	1
2	Resistor	10kΩ	2
3	capacitor	0.1 MF	2
4	Connecting wires	-	As required
5	Trainer kit	XPD-ANADIGI	1
6	DSO		1
7	Patch cords	-	As required

SOFTWARE REQUIRED:- LT Spice XVII

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THEORY

An astable multivibrator operates as a free-running oscillator that switches periodically between two quasi-stable states without external triggering, delivering a continous rectangular wave output. The timing depends on the charging and discharging of the timing capacitor between the timer's interval lower trigger and upper threshold limits. In the high output state, the capacitor charges through both resistors in series yielding,

$$t_c = 0.69(R_A + R_B)C$$

Upon reaching the threshold level, the output switches low and enables the internal discharge path, allowing the capacitor to discharge through the second resistor alone,

$$t_d = 0.69 R_B C$$

The cycle repeats continously, giving a total period, $T = t_c + t_d$, an operating frequency of f and duty cycle of -

$$\text{Duty cycle} = \frac{t_c}{T} = \frac{R_A + R_B}{R_A + 2R_B}$$

PROCEDURE

i) The timer was placed and connected to DC Supply and ground.

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Circuit Diagram

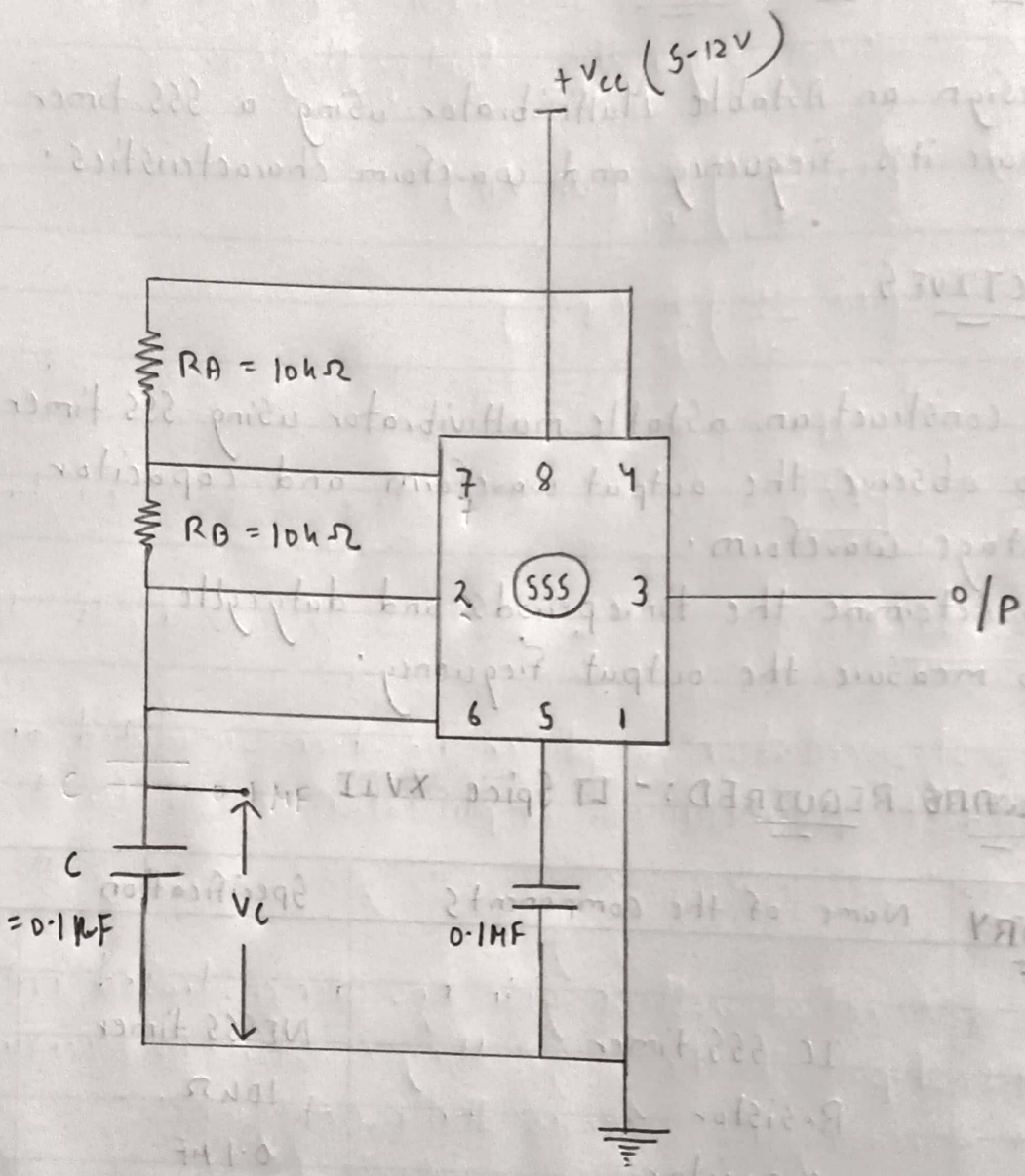


Fig1:- Circuit diagram of Astable Multivibrator using 555 timer IC

ii) The reset terminal was tied to the positive Supply rail.
iii) The first timing resistor was wired between Supply and discharge terminals.
iv) The second timing resistor was wired between discharge and threshold terminals.
v) The threshold and trigger terminals were shorted together.
vi) The timing capacitor was connected from the trigger terminal to ground.
vii) A bypass capacitor was tied from the control terminal to ground.
viii) Oscilloscope probes were placed at the output and capacitor terminals.
ix) The circuit was energized to display the output and capacitor waveforms.
x) Waveform time intervals were measured to determine pratical frequency and duty cycle.

CALCULATION

From the figure we can say,

$R_A = 10k\Omega$, $R_B = 10k\Omega$, $C = 0.1MF$

So, now for finding tc and td,

So, $t_c = 0.69(R_A + R_B)C$

$= 0.69 \times 20 \times 10^3 \times 0.1 \times 10^{-6}$

$= 1.38 \times 10^{-3}$

$= 1.38mS = 1380\mu S$

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Now, $T_d = 0.69 R_B C$

$= 0.69 \times 10 \times 10^3 \times 0.1 \times 10^{-6}$

$= 0.69 \times 10^{-3}$

$= 690\,\mu s$

Now, we can say,

$T(\text{theoretical}) = T_c + T_d$

$= 2070\,\mu s$

and $T(\text{pratical}) = 2050\,\mu s$

and also we can say, from simulation,

$T(\text{Simulation}) = 2080\,\mu s$

Now, we can calculate duty cycle as,

$$\text{Duty cycle} = \frac{R_A + R_B}{R_A + 2R_B} = \frac{10+10}{10+20} = \frac{20}{30} = 66.6\%$$

Now from the pratical observation,

Capacitor Voltage, $V_c(\text{pratical}) = 4.08\,V$

$V_c(\text{simulation}) = 3.96\,V$

and Similarly for o/p voltage,

$V_{o/p}(\text{pratical}) = 12.02\,V$

$V_{o/p}(\text{Simulation}) = 12\,V$

So, from this we can calculate % Error,

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So, for Capacitor Voltage,

$$\% \text{ Error} = \left| \frac{\text{Pratical} - \text{Simulation}}{\text{Simulation}} \right| \times 100\ \%.$$

$$= \frac{4.08 - 3.96}{3.96} \times 100 = \boxed{2.9\%}.$$

For o/p voltage,

$$\% \text{ Error} = \frac{12.02 - 12}{12} \times 100 = \boxed{0.17\%}.$$

For Time period we can say,

$$\% \text{ Error} = \left| \frac{2050 - 2070}{2070} \right| \times 100$$

(Here, $\% \text{ Error} = \frac{\text{Pratical} - \text{Theoretical}}{\text{Theoretical}} \times 100$)

$$= \boxed{0.96\%}$$

CONCLUSION

i) The astable multivibrator Circuit using 555 timer IC has been assembled, tested and Simulated to generate continous Square wave oscillations with independent charging and discharging paths.

ii) The theoretical time period of 2070 μs closely aligned with the pratical value of 2050 μs showing a minimal timing error of 0.96%.

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iii) The capacitor voltage and peak to peak output voltage have demonstrated close agreement between pratical and Simulated measurements with percentage errors of 2.9% and 0.17% respectively.

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